Merritt Losert

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Education

2018 - 2024 **University of Wisconsin-Madison**, PhD (Physics) (GPA: 3.82)
Advisors: Mark Friesen, Susan Coppersmith
Thesis: *Alloy Disorder, Valley Splitting, and Shuttling for Spin Qubits in Silicon/Silicon-Germanium Heterostructures*

2013 - 2017 **Dartmouth College**, BA, *Magna Cum Laude* (GPA: 3.81)
Majors: Physics, Computer Science. Minor: German Studies

Positions

National Institute of Standards and Technology (College Park, MD) Aug 2017 - Aug 2018
NRC Postdoctoral Fellow

Advisor: Justyna Zwolak. Machine learning and automation for semiconductor quantum dots.

Galin Education (Madison, WI) Sep 2019 - Sep 2024
Tutor

Tutoring high school and college students in math, physics, ACT, and SAT prep.

University of Wisconsin-Madison (Madison, WI) June 2019 - Dec 2024
Research Assistant

Simulations and theory for spin qubit devices, focusing on alloy disorder, valley splitting, and spin shuttling in Si/SiGe heterostructures.

University of Wisconsin-Madison (Madison, WI) Aug 2018 - Dec 2019
Teaching Assistant

Physics 103/104 (Introductory Physics) and ECE 532 (Matrix Methods in Machine Learning)

Alarm.com (Denver, CO) Aug 2017 - Aug 2018
Software Engineer

Full stack engineer (Sql Server, C#/.NET) working on ZWave devices, home automation technology and data analytics

Publications

[1] Róbert Németh, Vatsal K. Bandaru, Pedro Alves, Emma Brann, Owen M. Eskandari, Hudaiba Soomro, Avani Vivrekar, M.A. Eriksson, **Merritt P. Losert**, and Mark Friesen. "Omnidirectional Shuttling to Avoid Valley Excitations in Si/SiGe Quantum Wells". *PRX Quantum* 7 (2026), p. 010336. URL: https://link.aps.org/doi/10.1103/615j-xjyh.

[2] Yasuo Oda, **Merritt P. Losert**, and J. P. Kestner. "Suppressing Si Valley Excitation and Valley-Induced Spin Dephasing for Long-Distance Shuttling". *Phys. Rev. Lett.* 136 (2026), p. 020802. URL: https://link.aps.org/doi/10.1103/s624-dvvs.

[3] Benjamin D. Woods, **Merritt P. Losert**, Robert Joynt, and Mark Friesen. "g-Factor Theory of Si/SiGe Quantum Dots: Spin-Valley and Giant Renormalization Effects". *Phys. Rev. Lett.* 136 (20 May 2026), p. 206201. DOI: 10.1103/rd84-5g2x. URL: https://link.aps.org/doi/10.1103/rd84-5g2x.

[4] Jonathan C. Marcks, Emily Eagen, Emma C. Brann, **Merritt P. Losert**, Talise Oh, J. Reily, Christopher S. Wang, Daniel Keith, Fahd A. Mohiyaddin, Florian Luthi, Matthew J. Curry, Jiefei Zhang, F. Joseph Heremans, Mark Friesen, and M. A. Eriksson. "Valley splitting correlations across a silicon quantum well containing germanium". *Nature Communications* (2025). URL: https://doi.org/10.1038/s41467-025-67325-z.

[5] Lucas E. A. Stehouwer, **Merritt P. Losert**, Maia Rigot, Davide Degli Esposti, Sara Martí-Sánchez, Maximillian Rimbach-Russ, Jordi Arbiol, Mark Friesen, and Giordano Scappucci. "Engineering Ge Profiles in Si/SiGe Heterostructures for Increased Valley Splitting". *Nano Letters* 25.34 (Aug. 2025), pp. 12892–12898. URL: https://doi.org/10.1021/acs.nanolett.5c02848.

[6] Jan Klos, Jan Tröger, Jens Keutgen, **Merritt P. Losert**, Nikolay V. Abrosimov, Joachim Knoch, Hartmut Bracht, Susan N. Coppersmith, Mark Friesen, Oana Cojocaru-Mirédin, Lars R. Schreiber, and Dominique Bougeard. "Atomistic Compositional Details and Their Importance for Spin Qubits in Isotope-Purified Silicon Quantum Wells". *Advanced Science* 11.42 (2024), p. 2407442. URL: https://onlinelibrary.wiley.com/doi/abs/10.1002/advs.202407442.

[7] **Merritt P. Losert***, Max Oberländer*, Julian D. Teske, Mats Volmer, Lars R. Schreiber, Hendrik Bluhm, S.N. Coppersmith, and Mark Friesen. "Strategies for Enhancing Spin-Shuttling Fidelities in Si/SiGe Quantum Wells with Random-Alloy Disorder". *PRX Quantum* 5 (4 Nov. 2024), p. 040322. URL: https://link.aps.org/doi/10.1103/PRXQuantum.5.040322.

[8] **Merritt P. Losert**, M. A. Eriksson, Robert Joynt, Rajib Rahman, Giordano Scappucci, Susan N. Coppersmith, and Mark Friesen. "Practical strategies for enhancing the valley splitting in Si/SiGe quantum wells". *Phys. Rev. B* 108 (2023), p. 125405. URL: https://link.aps.org/doi/10.1103/PhysRevB.108.125405.

[9] J. P. Dodson, H. Ekmel Ercan, J. Corrigan, **Merritt P. Losert**, Nathan Holman, Thomas McJunkin, L. F. Edge, Mark Friesen, S. N. Coppersmith, and M. A. Eriksson. "How Valley-Orbit States in Silicon Quantum Dots Probe Quantum Well Interfaces". *Phys. Rev. Lett.* 128 (2022), p. 146802. URL: https://link.aps.org/doi/10.1103/PhysRevLett.128.146802.

[10] Thomas McJunkin, Benjamin Harpt, Yi Feng, **Merritt P. Losert**, Rajib Rahman, J. P. Dodson, M. A. Wolfe, D. E. Savage, M. G. Lagally, S. N. Coppersmith, Mark Friesen, Robert Joynt, and M. A. Eriksson. "SiGe quantum wells with oscillating Ge concentrations for quantum dot qubits". *Nat. Commun.* 13 (2022), p. 7777. URL: https://doi.org/10.1038/s41467-022-35510-z.

[11] Brian Paquelet Wuetz*, **Merritt P. Losert***, Sebastian Koelling*, Lucas E. A. Stehouwer, Anne-Marije J. Zwerver, Stephan G. J. Philips, Mateusz T. Mądzik, Xiao Xue, Guoji Zheng, Mario Lodari, Sergey V. Amitonov, Nodar Samkharadze, Amir Sammak, Lieven M. K. Vandersypen, Rajib Rahman, Susan N. Coppersmith, Oussama Moutanabbir, Mark Friesen, and Giordano Scappucci. "Atomic fluctuations lifting the energy degeneracy in Si/SiGe quantum dots". *Nat. Commun.* 13 (2022), p. 7730. URL: https://doi.org/10.1038/s41467-022-35458-0.

[12] Brian Paquelet Wuetz, **Merritt P. Losert**, Alberto Tosato, Mario Lodari, Peter L. Bavdaz, Lucas Stehouwer, Payam Amin, James S. Clarke, Susan N. Coppersmith, Amir Sammak, Menno Veldhorst, Mark Friesen, and Giordano Scappucci. "Effect of Quantum Hall Edge Strips on Valley Splitting in Silicon Quantum Wells". *Phys. Rev. Lett.* 125 (2020), p. 186801. URL: https://link.aps.org/doi/10.1103/PhysRevLett.125.186801.

Preprints

[1] **Merritt P. R. Losert**, S. N. Coppersmith, and Mark Friesen. *Using a spin-triplet encoding to enhance shuttling fidelities in Si/SiGe quantum wells.* 2026. arXiv: 2605.12687.

[2] Collin C. D. Frink, Talise Oh, E. S. Joseph, **Merritt P. Losert**, E. R. MacQuarrie, Benjamin D. Woods, M. A. Eriksson, and Mark Friesen. *Reducing strain fluctuations in quantum dot devices by gate-layer stacking.* 2025. arXiv: 2312.09235.

[3] **Merritt P. R. Losert**, Dario Denora, Barnaby van Straaten, Michael Chan, Stefan D. Oosterhout, Lucas Stehouwer, Giordano Scappucci, Menno Veldhorst, and Justyna P. Zwolak. *Automated electrostatic characterization of quantum dot devices in single- and bilayer heterostructures.* 2025. arXiv: 2601.00067.

[4] **Merritt P. R. Losert**, Utkan Güngördü, S. N. Coppersmith, Mark Friesen, and Charles Tahan. *The effects of alloy disorder on strongly-driven flopping mode qubits in Si/SiGe.* 2025. arXiv: 2512.19658.

[5] Benjamin D. Woods, **Merritt P. Losert**, Nasir R. Elston, M. A. Eriksson, S. N. Coppersmith, Robert Joynt, and Mark Friesen. *Statistical characterization of valley coupling in Si/SiGe quantum dots via g-factor measurements near a valley vortex.* 2025. arXiv: 2507.05160.

In prep (pdfs available upon request)

[1] **Merritt P. Losert**, Rajib Rahman, Lars R. Schreiber, S. N. Coppersmith, and Mark Friesen. *Using valley hot spots to enhance spin-shuttling fidelity in Si/SiGe heterostructures.*

* *denotes equal contribution*

Patents

US Patent No. 12439724 — Granted 2025

"Silicon-Germanium alloy-based quantum dots with increased alloy disorder and enhanced valley splitting"

US Patent Application No. 18/975,615 — Filed 2024

"Two-dimensional conveyor-mode spin qubit shuttling devices"

Fellowships and Awards

NIST NRC Postdoctoral Fellowship — 2025

LQC QuaCR Graduate Fellowship — 2022